Template test: damped harmonic oscillator

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Abstract

This document is a working example that checks every part of the template. Each section says what it tests and what you should see in the PDF if that part works. Once the checks pass, delete the files in sections/ and the example figure, and start writing.

1 Introduction

This section checks the basic text features. A paragraph is written as ordinary continuous text, and a single line break in the source does not change anything in the PDF, because LaTeX only starts a new paragraph at a blank line.

A second paragraph should appear after a blank space and without indentation, which shows that the parskip package is active. The first page should have no header, a rule above the footer, and the three author emails at the bottom left. Clicking a symbol next to an author name should jump to the corresponding email, and clicking an email should open your mail program. From the second page on, the header should show the assignment on the left and the course on the right, with a thick rule below.

The bibliography works if the following citations appear as numbers in brackets and not as question marks. Hawking showed that black holes emit thermal radiation [1], and the classical theory of the oscillator used below can be found in any electrodynamics or mechanics text [2, 3]. The citation links should be dark red.

Cross-references work if the following sentence contains working blue links with the right names and numbers. The model is derived in Section 2, its solution is Eq. (8), and the results are in Fig. 1 and Table 1.

Notes between co-authors appear as colored boxes inside the text:

This is an example note. To hide every note, add the disable option to todonotes at the top of main.tex.

External links should be red, for example the TeX Live home page.

2 Theory

This section checks numbered equations, the custom commands from the preamble, and the mathematical packages.

Let $x\colon \mathbb{R} \to \mathbb{R}$ be the position of a mass m attached to a spring of constant k and subject to a friction force proportional to the velocity, with coefficient b. Newton's second law gives

$$m\ddot{x} + b\dot{x} + kx = 0. \tag{1}$$

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Dividing by m and defining

$$\gamma \coloneqq \frac{b}{2m}, \qquad \omega_0 \coloneqq \sqrt{\frac{k}{m}}, \tag{2}$$

Eq. (1) becomes

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0. \tag{3}$$

We look for solutions of the form $x(t) = e^{rt}$ with $r \in \mathbb{C}$. Then $\dot{x} = re^{rt}$ and $\ddot{x} = r^2 e^{rt}$, and substituting into Eq. (3) gives

$$\left(r^2 + 2\gamma r + \omega_0^2\right) e^{rt} = 0. \tag{4}$$

Since $e^{rt} \neq 0$ for every t, the exponent must satisfy the characteristic equation

$$r^2 + 2\gamma r + \omega_0^2 = 0, \tag{5}$$

whose roots are

$$\begin{aligned} r_\pm &= \frac{-2\gamma \pm \sqrt{4\gamma^2 - 4\omega_0^2}}{2} \\ &= -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \end{aligned} \tag{6}$$

In the underdamped case, $\gamma < \omega_0$, the square root is imaginary. Writing $\gamma^2 - \omega_0^2 = -\omega_d^2$ with

$$\omega_d \coloneqq \sqrt{\omega_0^2 - \gamma^2} > 0, \tag{7}$$

the roots are $r_\pm = -\gamma \pm i\omega_d$. The general real solution is a combination of $e^{r_+ t}$ and $e^{r_- t}$, which can be written as

$$x(t) = A\, e^{-\gamma t} \cos\left(\omega_d t + \varphi\right), \tag{8}$$

where the amplitude A and the phase φ are fixed by the initial conditions. For $x(0) = x_0$ and $\dot{x}(0) = 0$, Eq. (8) gives $x_0 = A\cos\varphi$, and differentiating,

$$\dot{x}(0) = -A\gamma\cos\varphi - A\omega_d \sin\varphi = 0 \quad \Longrightarrow \quad \tan\varphi = -\frac{\gamma}{\omega_d}. \tag{9}$$

The commands `\R`, `\N` and `\sub` defined in the preamble should render as $\mathbb{R}$, $\mathbb{N}$ and $\mathbb{N} \subset \mathbb{R}$, and `\mathscr` as $\mathscr{L}$. Quantities with units are written with siunitx, for example a period of $2.0\,\mathrm{s}$ or an acceleration of $9.81\,\mathrm{m\,s^{-2}}$.

The physics and braket packages are checked with the quantum version of the same oscillator. Its Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega_0^2 \hat{x}^2, \tag{10}$$

its energies form the set $\{\, E_n \mid n \in \mathbb{N} \,\}$ with $E_n = \hbar\omega_0\left(n + \tfrac{1}{2}\right)$, and a normalized state $|\psi\rangle$ satisfies $\langle\psi|\psi\rangle = \int |\psi(x)|^2 \,\mathrm{d}x = 1$. The expectation value written with physics, $\langle\psi|\hat{H}|\psi\rangle$, and with braket, $\langle\psi|\hat{H}|\psi\rangle$, should look identical.

Starting from $[\hat{x}, \hat{p}] = i\hbar$, and since $\hat{x}$ commutes with the potential term of Eq. (10),

$$\begin{aligned} \left[\hat{x}, \hat{H}\right] &= \frac{1}{2m}\left[\hat{x}, \hat{p}^2\right] \\ &= \frac{1}{2m}\left(\hat{p}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{p}\right) \\ &= \frac{1}{2m}\left(i\hbar\,\hat{p} + i\hbar\,\hat{p}\right) = \frac{i\hbar}{m}\,\hat{p}. \end{aligned} \tag{11}$$

The Heisenberg equation then gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{x}\rangle = \frac{1}{i\hbar}\left\langle\left[\hat{x},\hat{H}\right]\right\rangle = \frac{1}{i\hbar}\frac{i\hbar}{m}\langle\hat{p}\rangle = \frac{\langle\hat{p}\rangle}{m}, \tag{12}$$

the quantum counterpart of $\dot{x} = p/m$, and partial derivatives such as $\frac{\partial\psi}{\partial x}$ and $\frac{\partial^2\psi}{\partial x^2}$ should appear as fractions with ∂.

3 Figures and tables

This section checks figures from the figures/ folder, subfigures, tables and multiple columns.

Figure 1 is a PDF file in figures/, produced by the Python script next to it, figures/damped-oscillator.py. If it appears, the graphics path works. Its label should read Fig. in bold followed by a period.

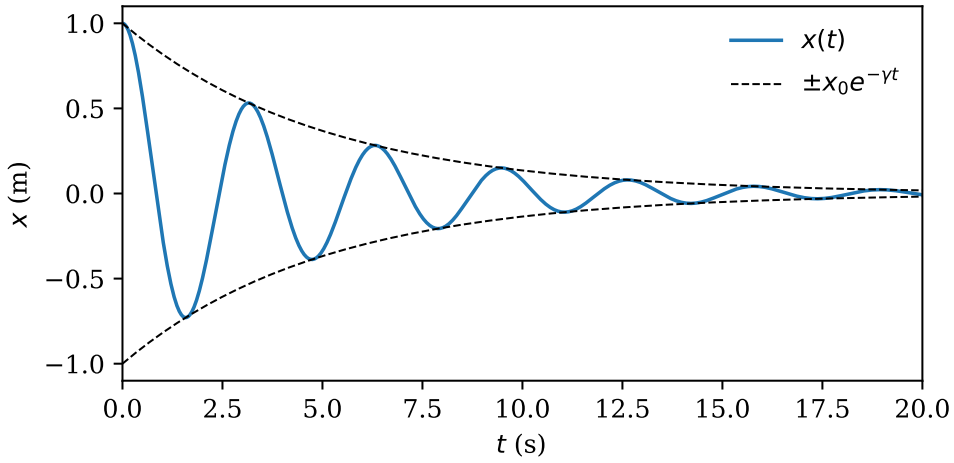


Fig. 1. Position of the damped oscillator, Eq. (8), with the parameters of Table 1. The dashed lines are the envelope $\pm x_0 e^{-\gamma t}$.

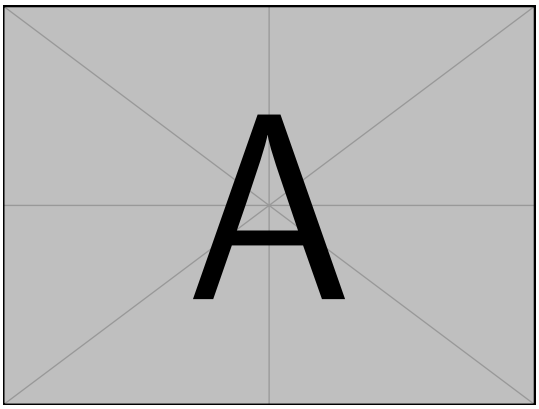
Figure 2 has two subfigures that can be referenced separately, as Fig. 2a and Fig. 2b. The images are placeholders that come with TeX Live.

Table 1 uses booktabs rules and siunitx columns, so the numbers are aligned on the decimal point.

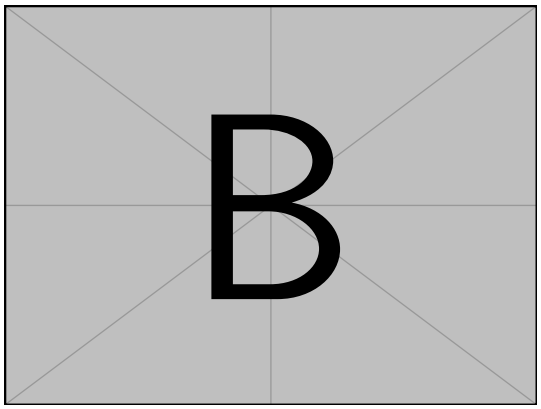
Table 1. Parameters used in Fig. 1.

Quantity	Symbol	Value	Unit
Mass	m	1.0	kg
Spring constant	k	4.0	N/m
Damping coefficient	b	0.4	kg/s
Initial position	x_0	1.0	m
Damped angular frequency	ω_d	1.990	rad/s

The last check is the multicol package. The following paragraph should be split into two columns.



(a) First subfigure.



(b) Second subfigure.

Fig. 2. Two subfigures in one figure.

With $m = 1.0\,\mathrm{kg}$, $k = 4.0\,\mathrm{N/m}$ and $b = 0.4\,\mathrm{kg/s}$, Eq. (2) gives $\omega_0 = 2.0\,\mathrm{rad/s}$ and $\gamma = 0.2\,\mathrm{s}^{-1}$. Then Eq. (7) gives $\omega_d = \sqrt{4.00 - 0.04}\,\mathrm{rad/s} = \sqrt{3.96}\,\mathrm{rad/s} \approx 1.990\,\mathrm{rad/s}$. The amplitude decays by a factor e in a time $1/\gamma = 5\,\mathrm{s}$, which is the scale visible in Fig. 1. If this text is in two balanced columns, everything in the template works.

References

[1] Stephen W. Hawking. “Particle creation by black holes”. In: *Communications in Mathematical Physics* 43.3 (1975), pp. 199–220. DOI: 10.1007/BF02345020.

[2] John David Jackson. *Classical Electrodynamics*. 3rd ed. New York: Wiley, 1999.

[3] Herbert Goldstein, Charles Poole, and John Safko. *Classical Mechanics*. 3rd ed. San Francisco: Addison-Wesley, 2002.